ZOO403 Midterm Important Mega Subjective File

All Short & long Questions

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1) What is associative behavior? 2 marks

Associative Learning "A learning that involves making associations between experiences"
Example
• A mouse bites a brightly colored caterpillar

• It tastes bad

• Mouse avoids attacking insects of similar color in future.

Associative Learning
• Ability to associate one environmental feature (such as a color) with another (such as a foul taste)

2) Two examples of morphological response .2 marks

Example:
• Chameleon change color when it feel threatened.

• Arctic hare change its coat color from blue-grey to white for better camouflage during winters and turn again to blue-grey in spring

3) What is learned behavior? 3 marks

Thorpe defined learning as a long lasting adaptive change in behavior resulting from past experience; hence it is acquired during the life time of an individual
Learned behaviors

• modified by experience

• variable, changeable

• flexible with a complex & changing environment

4) Fixed Action pattern. 3 marks

• It is a sequence of unlearned, innate behavior that is unchangeable.

• Once initiated; it is usually carried to completion.

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• It is hardwired, however, not purely genetic, may improve with experience.
• FAP is triggered by an external sensory stimulus known as “sign stimulus”.

5) Explanation of innate behavior with example. 5 marks

i. automatic, fixed, “built-in”, no “learning curve”

ii. despite different environments, all individuals exhibit the behavior

iii. Examples: Early survival, reproduction, kinesis, taxis.

Examples

- **Web making in spiders**
- **Nest building in birds**
- **Fighting among male stickleback fish**

6). Imprinting. 5 marks

“Formation of a long-lasting behavioral attachment to a particular individual or object at a specific stage in life”

Imprinting

• A type of learning behavior

• A very young animal develops attachment with the very first object it has visual, auditory or tactile experience

• It thereafter follows that object

Example

• Seagulls

• Sensitive period for a parent to bond with its young lasts one to two days

• If bonding does not occur, parent will not care for the infant, leading to death of the offspring

Imprinting—Example

• Classic experiment of Konrad Lorenz with geese

• Konrad Lorenz made goslings to imprint on him

• Goslings followed him as though he was their mother

7).Habitat of dolphin. (2)

Dolphins are found all over the world; generally in shallow cold waters as well as the warm tropical waters

8).Kinesis (2)

Kinesis: “a change in activity or turning rate in response to a change in location”.

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Kinesis—Example:

• Kinesis in sow bugs in response to variation in humidity.

• Move more rapidly in dry areas and become slow in humid areas.

• They survive better in moist areas.

9). Enlist the type of learning. (3)

- Habituation
- Imprinting
- Classical conditioning
- Trial and error learning
- Latent learning
- Insight
- Reasoning
- Cognition

10)... Characteristics of learned behaviors.(3)

- Acquired during the life of an organism due to constant experience
- Experience dependent and can be modified through experience
- Learning is flexible
- Learning behavior differs from individual to individual, to individual among the same species hence not species specific
- Learned behavior is certainly not inherited, though the ability to learn is almost certainly inherited as it is dependent on the development of the nervous system of the organism which is inherited.
- All organisms have the ability to learn at least to some extent.
- However learning in humans surpasses all other animals. Almost everything we do has been learned. No other species surpasses the humans in their amount and range of information that is acquired through learning.

11). Note on experiment of Angelo d'Arrigo. (5)

Angelo d'Arrigo extended the technique used by Konrad Lorenz.

• He noted that the flight of hang-glider is very similar to the flight patterns of migratory birds.

• Both use updrafts of thermal currents to gain altitude that then permits soaring flight over distance.

• He used this method to reintroduce threatened species of raptors.

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• As birds hatched in captivity have no mentor birds to teach them traditional migratory routes, D'Arrigo hatched chicks under the wing of his glider and they imprinted on him. Then, he taught the fledglings to fly and to hunt.

• The young birds followed him not only on the ground (as with Lorenz) but also in the air as he took the path of various migratory routes

12). Characteristics of innate behaviour.(5)

• Pattern of behavior is inherited. It is passed on from parents to offspring
• It is unlearned behavior.

• It occurs in all the members of a species hence it is species specific and predictable.

• It is not dependent on past experience as it is an inborn response to a stimulus.

• It takes place in individuals even when kept in isolation away from their fellow members.

• Innate behavior has high adaptability and survival value

13).Animal orientation long question

Animal Orientation: "Ability of an animal to determine its position in space and move predictably with respect to specific stimuli"
Animal Orientation: Examples:
• Standing up after falling down (orientation against gravity)

• Orienting towards incoming light

• Acoustic orientation

• Olfactory orientation
Sensory Motor Basis of Orientation:
• Animal orientation is a complex process.

• Requires integration of sensory input and coordination of motor output.
Input-Output Integration:
Integration includes:
• Receiving information through sensory receptor neurons.

• Processing and correlating it in the CNS.

• Forming a response through muscles that cause orienting movement

14).Spatial learning

Spatial Learning "Capacity to establish a memory that reflects environment's spatial structure"
Spatial Learning
• Spatial learning enhances an organism's fitness in environment

• Every natural environment has some spatial variation e.g.,

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• Locations of nest sites

• Hazards

• Food

• Prospective mates

Spatial Learning—Example

• Behavior of a female digger wasp

• Nests in small burrows dug into sand mounds

• A wasp covers the entrance of her nest with sand when going on hunting

• Upon return, she flies directly to her hidden nest

15).Anolis lizard hormone

Anolis Lizards: Testosterone is critical for stimulating male sexual behaviors. Castration reduces the display of male reproductive behaviors, and testosterone can prevent or reverse that effect.

• Green Anoles use their pink dewlaps for two purposes.

❖ For establishing territory among adult males.

❖ By displaying his dewlap, the lizard can tell other males that they have entered his home range.

16).Chemical signals advantages

• Provide a simple message that can last for hours or days

• Are equally effective during night or day

• May be transported over long distances by air

• Take relatively little energy to produce

17).Crayfish controlling factor??

18).What is taxis?

Taxis: "An oriented movement toward (positive taxis) or away (negative taxis) from some stimulus".

Taxis—Examples:

• River fishes (e.g. trout) swim or orient in an upstream direction (toward the current).

• Helps keep the fish from being swept away.

Or

Taxis: Taxis is the simplest type of innate or stereotyped behavior. It is an orientation of an animal (directed either towards or away) in response to the source of stimulus.

Types:

1. Positive taxis If the orientation is towards the stimulus it is called as positive taxis.

2. Negative taxis If the orientation is away from the stimulus it is known as negative taxis.

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1. Positive taxis

- A plant or animal that move towards light shows positive photo taxis.
- The protozoan, Euglena response to a variety of stimuli and is very sensitive to light. It swims towards light; hence it shows positive photo taxis.

2. Negative taxis

- • Certain burrowing animals more away from the source of light exhibits negative photo taxis.
- Maggots that are about to pupate moves away from light source towards a dark location hence they show negative photo taxis

19).Crayfish controlling factor long

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20).Types of innate behavior

There are different types of innate or stereotyped behavior

- taxis

ii. Irritability

iii. Kinesis

iv. Instincts

v. Reflexes

vi. Motivation

21).Behavior endocrinology

The scientific study of the interaction between hormones and behavior is called **behavioral endocrinology.** This interaction is bidirectional: hormones can influence behavior, and behavior can sometimes influence hormone concentrations.

22).Twins study

Influence of genetics and environment on human behavior, is explored by twin studies

- Behaviors of identical twins are compared which are:
- Raised apart
- Raised in the same household

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23).Difference between morphological and behavioral responses

1. Morphological Responses
- changing in the of the body appearance
- Growing thicker fur or change in fur color in winter

(Question 24)Examples:

• Chameleon change color when it feel threatened.

• Arctic hare change its coat color from blue-grey to white for better camouflage during winters and turn again to blue-grey in spring.

2. Behavioral Responses
- changing behavior to adapt to the change
- moving to a more favorable location
- Cooperative behavior
- Agonistic behavior when threatened

Examples:

• During the winter season, **Houbara bustard** finds refuge in plain lands of Sindh, Southern Punjab and Baluchistan.

25).Objectives of Ethology

• Stimuli that elicit behavior

• Physiological mechanisms mediating response

• Animal's experiences that influence behavior

• Role of behavior in survival & reproduction

• Evolutionary history of behavior

26). Names of Organisms in which Researchers examined Cognition

Researchers have examined animal cognition in:

- ✦ mammals (primates, cetaceans, elephants, dogs, cats, horses, cattle, raccoons and rodents),
- ✦ birds (including parrots, fowl and pigeons),
- ✦ (lizards, snakes, and turtles)
- ❖ Fishes
- ❖ Reptiles

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✦ Invertebrates (cephalopods)

27).Significance of Habituation

• An animal conserves energy and time by habituation to unimportant stimuli

• Energy and time are better spent on other important functions

28).Definition of Cognition

Cognition "The process of intellect, represented by awareness, understanding, reasoning, recalling memories and judgments"

Cognition

• Most complex form of learning

• Characteristically exhibited by primates

• Also shown by:

• Many mammals

• Cephalopods

• Insects

29).Biological Clocks

Animals respond rhythmically to cyclic fluctuations in environment e.g.

• Changes in seasons

• Daily light and dark periods

Rhythmic behaviors are controlled by biological clocks

• Biological clocks regulate daily and seasonal cyclic activities

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• Clocks have a molecular genetic basis

• Genes for time sense have been identified in many animals (human beings too)

Biological clocks may be:
• Circadian clocks

• Circannual clocks

• Lunar clocks

These clocks are influenced by:
• Dark and light periods

• Lunar cycles

30).Ethology

"Scientific study of animal behavior in natural environment

31).Imprinting

"Formation of a long-lasting behavioral attachment to a particular individual or object at a specific stage in life"

32).Nest Making

Nest building in birds_ an examples of instinctive or innate behavior
• A bird nest is the spot in which a bird lays and incubates its eggs and raises its young.

• For most birds, when a male and female form a breeding pair nest building activities will follow.

• Nest building is an instinctive behavior as birds have innate ability to build nests.

Nest Building Behavior:
• A **nest site** is first selected and nest building begins.

• This can involve both birds as a pair or either sex alone.

• Different species of birds build different nests. Some nests are very simple constructions, whilst others are more complex.

• Birds usually only build nests for holding their eggs and rearing chicks.

33).Ultimate causes

Ultimate Causes

We ask questions like
• Why animal behaves as it does?

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• What is the evolutionary cause of behavior?

These questions explain the ultimate cause of behavior. The ultimate level of causation refers to:

• The behavior's evolutionary significance

• How the behavior enhances reproductive fitness

34).Neuron components

Neuron has three components

- ❖ Cell body
- ❖ Axon(moving nerve impulse away from cell body)
- ❖ Dendron(moving nerve impulse toward the cell body)

35).Neuronal plasticity

• "Capacity of nervous system to change with experience"

• Basic architecture of nervous system establishes during embryonic development

• Behavioral patterns modify with time, age and experience

• Premier importance for fitness and survival.

36).Male hormone

Testosterone

37).Cocoon building insects

• Butterflies and moths are the most common cocoon building insects.

• Their larvae, which are **caterpillars**, are voracious eaters.

• Caterpillars spin silk, and this silk is used to form the cocoon for the pupil stage of development – the final stage before adulthood.

38).Stickleback Fish

• Sticklebacks exhibit stereotyped instinctive behavior such as migration, nest building, territoriality, aggression, courtship and parental care.

• They are a good example of male dominating and nesting behavior too.

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• Normally breed in late March and early August.

• Usually live in schools.

• But at the beginning of breeding season male isolate itself.

The reproductive behavior in three-spine stickleback fish: A sexually mature stickleback fish male will have a red colored belly to attract the female. A mature stickleback fish migrates to warm shallow waters

39).Orthokinesis

Ortho-kinesis: It is a response that involves changes in the speed of movement of the whole body in response to stimuli like humidity, preserve and diffused light.

Examples:

• Wood louse, *Porcellerio scaber,* tend to lose water from their body fairly rapidly when exposed to low humidity.

40).Symbolic language of Honey bee

• A honeybee finds a food source close to the hive (less than 50 m away)

• Returns to its hive

• Moves in tight circles, moving its abdomen from side to side "round dance"

• Motivates other bees to leave the hive

• Follow her for nearby food

Symbolic Language of Honeybees

• When food source is farther

• "Waggle dance" is performed

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• Swings half-circle in one direction, a straight run, a half-circle swing in other direction

• Communicates direction and distance of source

41).Insight learning

Insight is immediate and clear learning

• understanding that takes place without overt trial-and-error testing

• Insight occurs when animals recognize relationships (or make novel associations between objects or actions) that can help them solve new problems

• A type of learning that uses reason, especially to form conclusions, inferences, or judgments, to solve a problem.

• It is based on advanced perceptual abilities such as thought and reasoning
Kohlar's work on chimpanzees is the experimental proof of insight learning.
• During experiment Köhler placed a banana outside the cage of a hungry chimpanzee

• Two sticks was given to Chimpanzee, but too short for pulling in the food but joinable to make a single stick of sufficient length

• First; chimpanzee tried to use only a stick to reach banana

• Then it tried to use one stick to push the other along to touch the banana.

• Later, it accidentally joined the sticks and observed the result and used it to retrieve the banana.

• In repeated experiments, the chimpanzee joined the two sticks and took the banana immediately

42). Reproductive Behavior in Ring Doves

Male and female ring doves coordinate their participation in:

- Courtship
- Nest building
- Care of young
- Incubation

43).Types of animal behavior

There are 2 basic types of behavior.

- • Innate behaviors or Inherent behavior.
- • Learned behaviors or acquired behavior.
- 1. Nature/innate [Instinct or genes determine behavior]
- 2. Nurture/learned [Experience and learning influence behavior]

44).Reflex action

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A reflex action thus involves the sense organs and nervous system.

A **reflex action** requires a **reflex arc** which consists of:

- A sensory organ (receptor)
- A sensory nerve (afferent nerve)
- The spinal cord or brain
- An intermediate motor or efferent nerve
- A motor organ or effector.

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